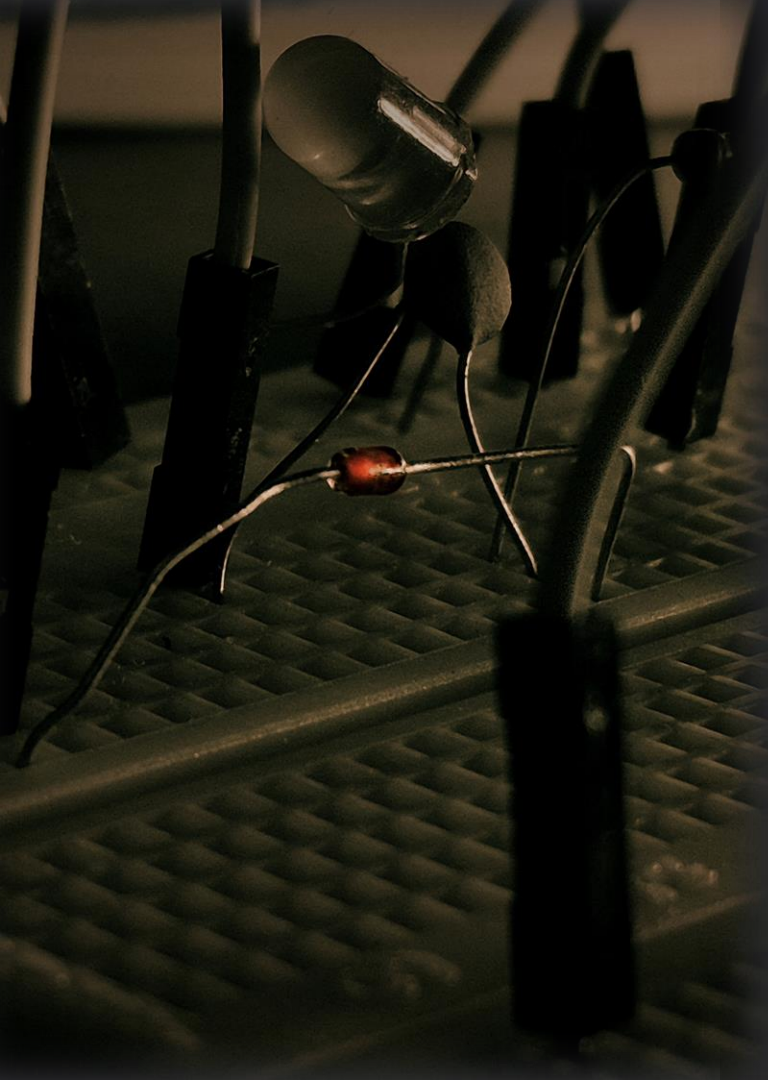




Quantum Random Number Generator (QRNG)

Real-Time Hardware-Based True Randomness Generation using
Quantum Noise

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Why Do We Need True Randomness?

The Problem with PRNGs

Most modern systems rely on **Pseudo Random Number Generators (PRNGs)** — algorithm-based sequences that are entirely predictable once the seed is known.

- Predictable under reverse-engineering attacks
- Critical security vulnerabilities in cryptographic use
- Pattern repetition over long sequences
- Insufficient entropy for key generation

Our Hardware Solution

A true ORNG exploiting the inherent unpredictability of quantum physics:



Quantum Tunneling Noise

Physical, non-deterministic source



Analog Amplification

LM358 op-amp signal conditioning



ESP32 Real-Time Sampling

Live ADC conversion & bit extraction

Objectives of the Project



True Quantum Randomness

Generate true random bits from physical quantum noise — not algorithmic approximation.



Signal Amplification

Amplify sub-millivolt noise signals to ADC-readable levels using LM358.



Real-Time Visualization

Stream and display live randomness data for immediate analysis.



Noise Capture

Capture weak quantum tunneling noise from a reverse-biased Zener diode.



Bit Extraction

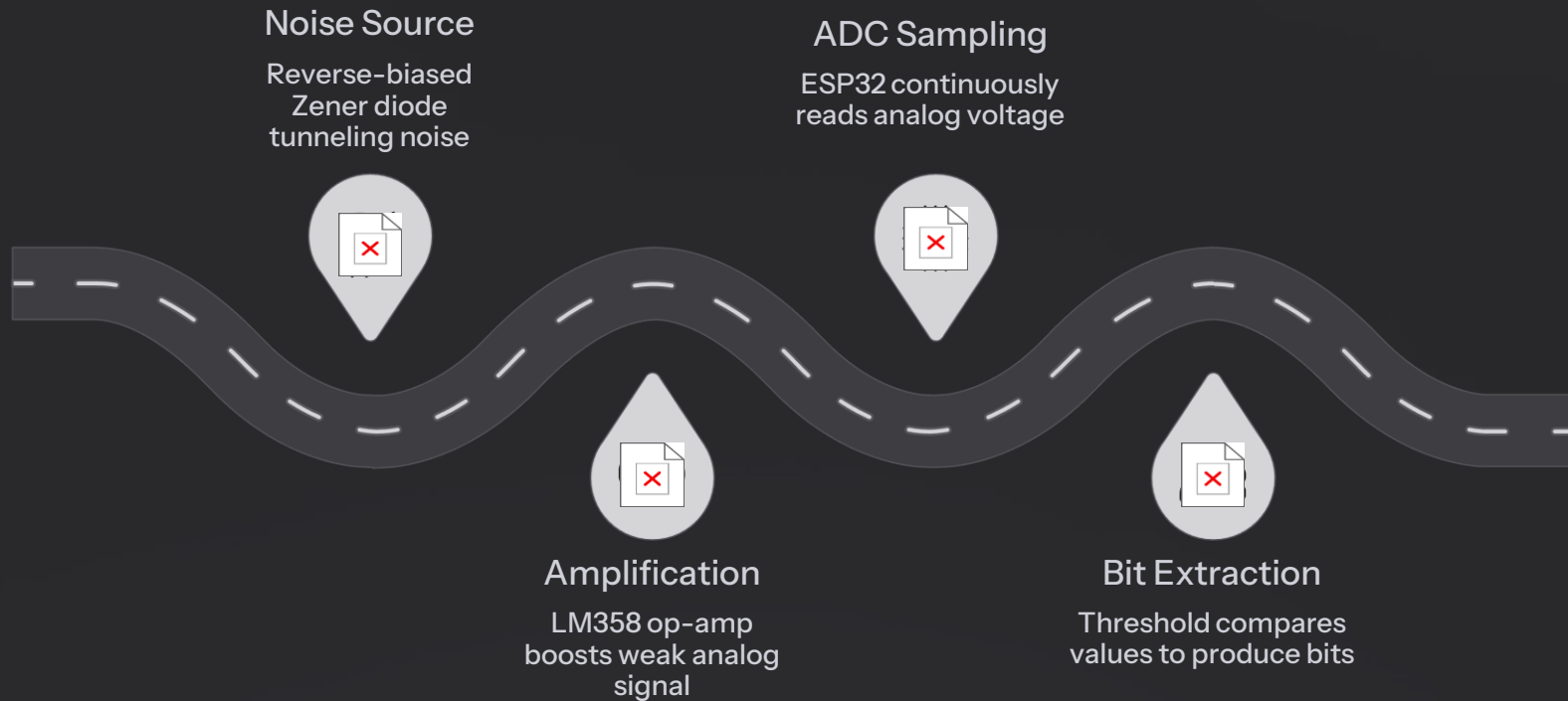
Convert amplified analog samples into a stream of digital 1s and 0s via threshold comparison.



Randomness Quality Analysis

Evaluate output quality using scatter plots, bit matrices, and serial plotter waveforms.

Working Principle



Each stage is carefully designed to preserve the physical entropy introduced at the quantum noise source, ensuring the final bit stream inherits true non-determinism.

Hardware Architecture

Key Components

LM358 Op-Amp

Two-stage amplification of weak noise signal

Potentiometer

Fine-tunes gain and bias for balanced 0/1 output

Zener Diode

Quantum noise source — operates in reverse breakdown

LM393N

Converts amplified analog quantum noise into clean digital 0/1 bitstream.

 Low-cost, portable, and fully adjustable — no external clock or seed required.

Real-Time Serial Plotter

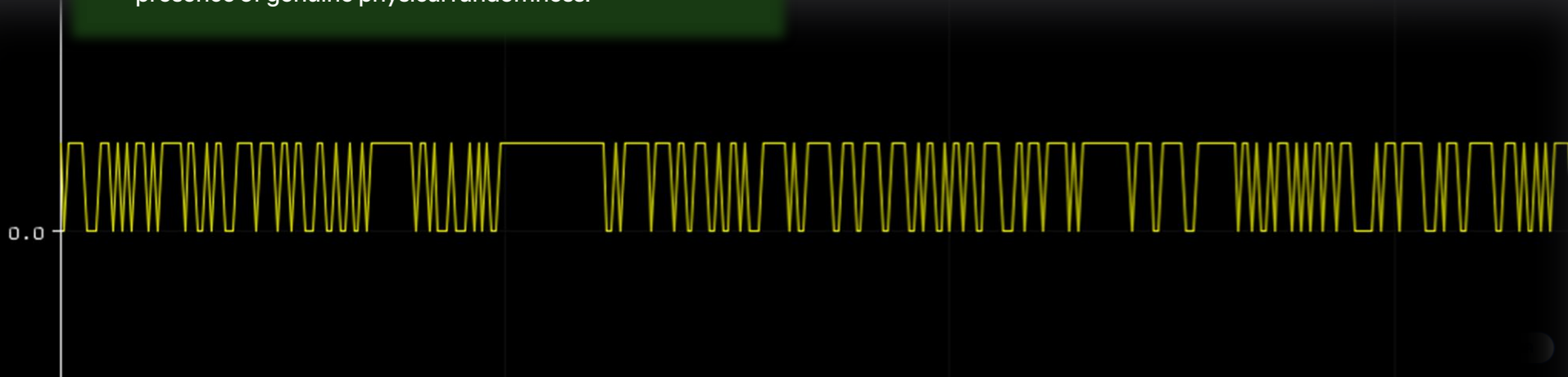
Purpose & Method

The ESP32 streams live ADC samples over USB serial. The Arduino Serial Plotter renders the analog noise waveform in real time, providing immediate visual confirmation of signal health.

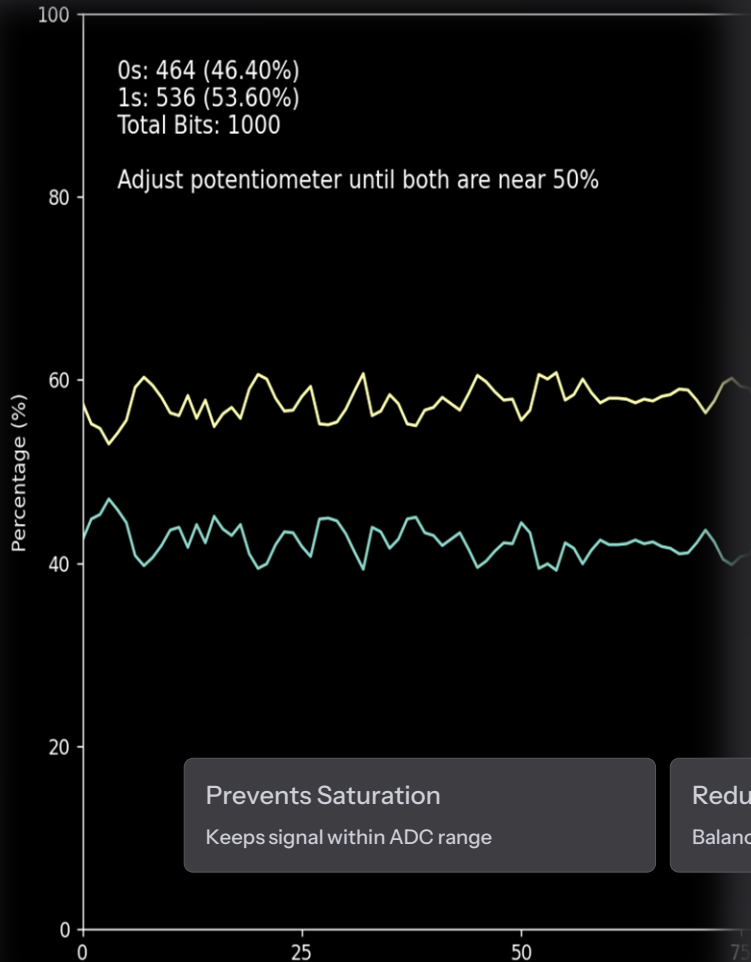
✓ Confirms signal stability, amplification quality, and presence of genuine physical randomness.

Key Observations

- **Irregular waveform** — no discernible periodicity
- **No repeating pattern** — amplitude changes unpredictably each sample
- **Random amplitude fluctuations** — consistent with physical quantum noise



Potentiometer Tuning & Bias Adjustment



Why Tuning Matters

Even a slight DC offset in the amplifier chain can skew the threshold comparator, producing more 1s than 0s (or vice versa). The potentiometer corrects this in real time.

The 0/1 ratio is monitored live and adjusted until statistical balance is achieved.

Prevents Saturation

Keeps signal within ADC range

Reduces Bit Bias

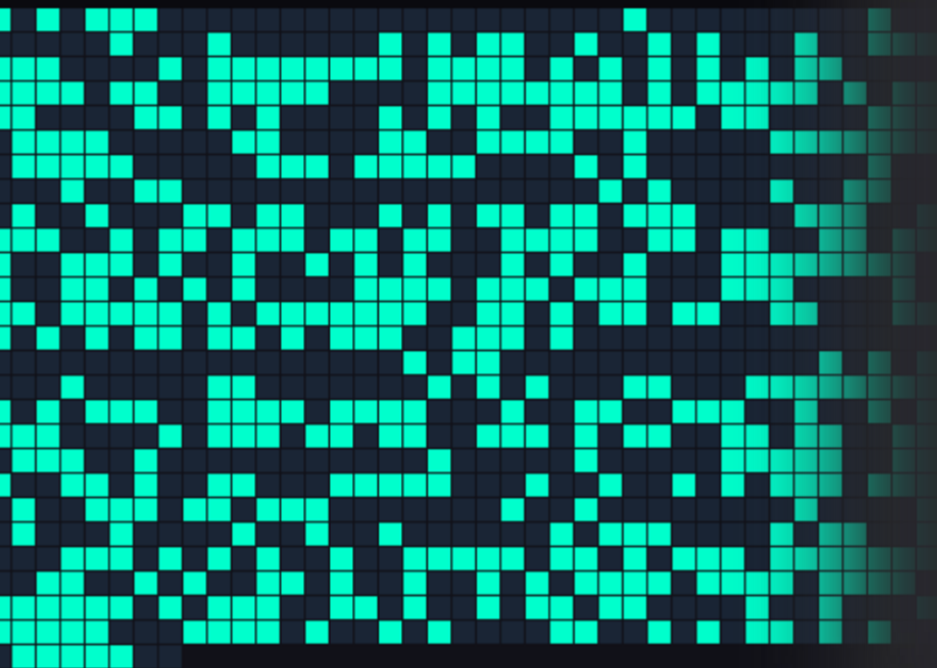
Balances 0/1 ratio toward 50%

Improves Sensitivity

Maximizes usable ADC resolution

Live Bit Stream

Each cell = 1 bit · cyan = 1 · dark = 0



Bit Matrix Visualization

Visual Randomness Inspection

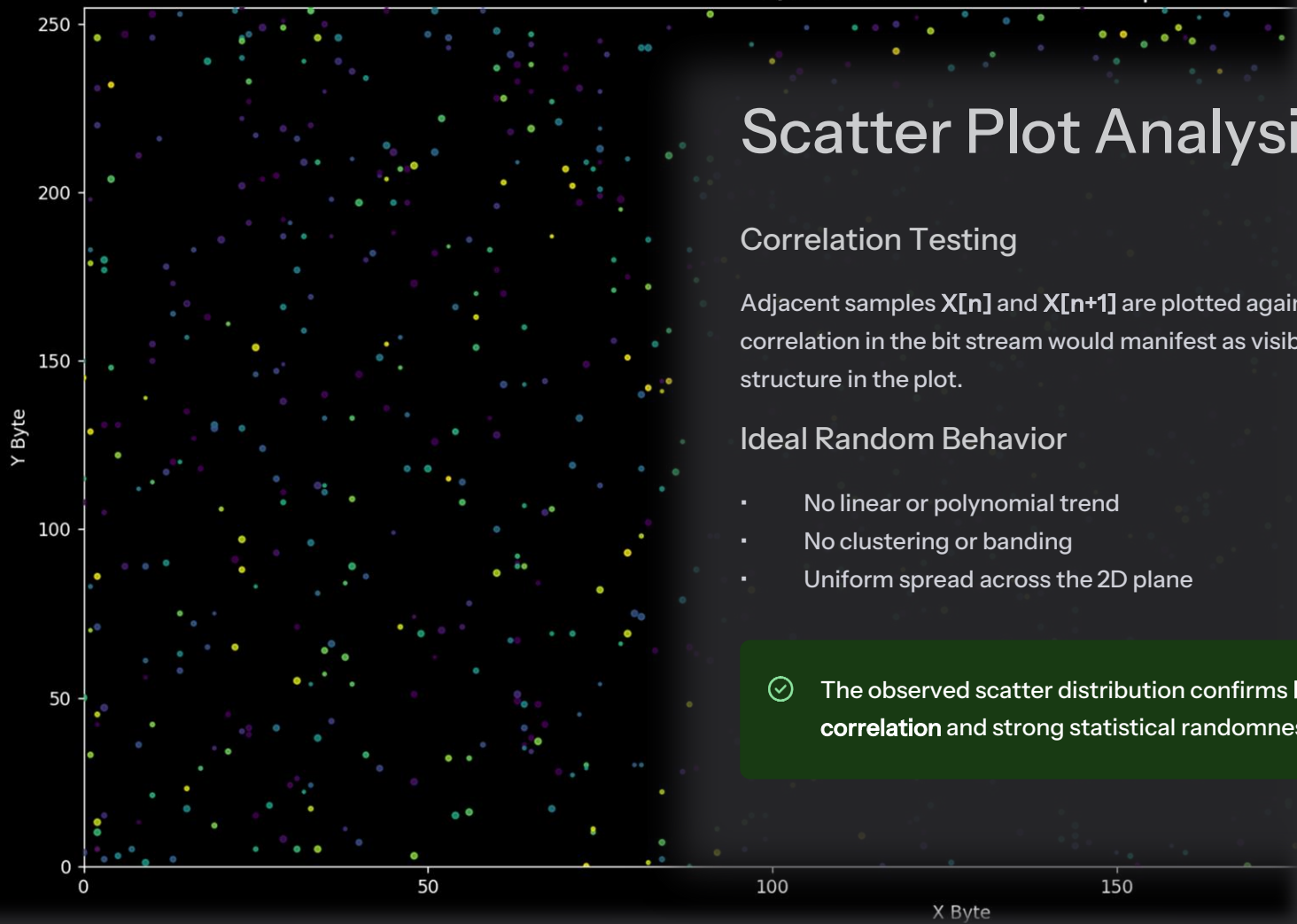
Generated bits are rendered as a 2D binary matrix: **white cells = 1**, **black cells = 0**. A truly random source produces a visually "noisy" matrix with no recognizable structure.

Ideal Characteristics

- No geometric structures or diagonal lines
- No repeated block patterns
- Approximately equal black/white density

✔ Observed matrix shows highly unpredictable distribution — consistent with a quantum entropy source.

1s: 1,071 0s: 1,225 ratio: 0.8743



Scatter Plot Analysis

Correlation Testing

Adjacent samples $X[n]$ and $X[n+1]$ are plotted against each other. Any correlation in the bit stream would manifest as visible clustering or linear structure in the plot.

Ideal Random Behavior

- No linear or polynomial trend
- No clustering or banding
- Uniform spread across the 2D plane



The observed scatter distribution confirms **low inter-sample correlation** and strong statistical randomness.

Applications, Results & Conclusion

Key Applications



Cryptography & Secure Comms

Unpredictable keys for encryption protocols



Quantum Key Distribution

Entropy source for QKD and post-quantum schemes



Scientific Simulations

Monte Carlo and stochastic modeling



Authentication Systems

Token generation and session nonces



Conclusions

- Successfully generated **true random bits** from quantum tunneling noise
 - Low-cost hardware: Zener diode + LM358 + ESP32
 - Bit matrix and scatter plot confirm **high-quality entropy**
 - Real-time visualization validates signal integrity end-to-end
 - Scalable architecture suitable for embedded security applications
-  QRNG provides a fundamentally stronger entropy source than any software-based PRNG.